

Question	Answer	Marks
5(a)	intensity $\propto$ (amplitude) ²	B1
5(b)(i)	$v = f\lambda$ or $c = f\lambda$	C1
	$f = 3.00 \times 10^8 / 0.060$ $= 5.0 \times 10^9 \text{ Hz}$	A1
5(b)(ii)	(at X path) difference = 3λ	M1
	(at X phase) difference = 0 or 1080°	M1
	so intensity is at a maximum/it is an intensity maximum	A1
5(b)(iii)	1. decrease in the distance between (adjacent intensity) maxima/minima	B1
	2. (intensity) maxima and minima exchange places	B1

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Question	Answer	Mark
6(c)	D 114	01